READSPACE

Mobile Application PRD, UI/UX Design & TRD

Version 1.0 · October 2026

Offline-first document reader · Optional Google Drive workspace · Android-first, iOS-ready

Product decision	Specification
Primary role	Mobile companion and standalone reader for local documents; cloud connection is optional.
Authentication	No login required to open local files, read offline, bookmark, highlight, or use local recent files.
Cloud access	User signs in and authorizes Google Drive only when they choose a cloud feature.
Offline principle	Local documents and cached/downloaded documents remain readable without internet. Drive browsing, refresh, and cloud transfer require connectivity.
Backend	Vercel API/backend with MongoDB for account-level metadata and secure Drive connection metadata; Google Drive stores cloud files.
Platforms	Android-first implementation, with an architecture and UX that can support iOS.
Design direction	Calm, polished reading-first interface; warm white/light surfaces, restrained indigo accent, comfortable typography, clear offline/cloud states.

1. Executive summary

ReadSpace Mobile is a native-feeling mobile document reader built around offline access. Users can browse and open files stored on their phone without creating an account. They can read supported documents, manage recent files, bookmarks, highlights and reading position locally. If they choose to connect Google Drive, they can browse cloud folders, open supported cloud documents, upload local files, download Drive files for offline reading, and resume cloud workspaces. Login and Drive authorization are optional and must never block local reading.

1.1 Product principles

- Local-first: local reading is the default and works without internet after the file is available on-device.
- Optional identity: no forced signup wall for local files.
- Explicit cloud consent: explain why Drive access is requested and request it only when the user taps Connect Google Drive or starts a cloud-only action.
- Honest offline behavior: show whether a file is local, downloaded for offline use, or cloud-only.
- Reader-first UI: content, typography and navigation matter more than toolbars.
- No silent destructive actions: uploads, downloads, replacement, deletion and conflict resolution require clear user intent.
- One renderer model: local and Drive documents use the same rendering and reading controls wherever the file format allows.

2. Goals and non-goals

2.1 Goals

- Open local documents using Android's system file picker and content URIs; support app-private files and downloaded files.
- Read supported PDF, TXT, Markdown, HTML and DOCX documents, with format-specific capabilities and graceful fallbacks.
- Keep a local recent-files list and restore reading positions after app restart.
- Support offline bookmarks, highlights and reader settings for supported formats.
- Let users connect Google Drive only when they want cloud access.
- Browse Drive folders, open supported files, upload selected local files, detect likely new files, and download cloud files to the device.
- Mark selected Drive documents as available offline by downloading a local copy.
- Protect each user's Google connection and cloud metadata from other accounts.
- Provide accessible touch targets, responsive layouts, dark theme, and clear network/sync status.

2.2 Non-goals for mobile MVP

- Full VS Code desktop-style multi-panel UI on a phone.
- A terminal, IDE, git integration or plugin marketplace.
- Real-time collaborative editing.
- Always-on bidirectional folder synchronization.
- Editing PDF or complex DOCX layouts.
- Unrestricted background scanning of all phone storage.
- Using a user's Google API key as a substitute for OAuth authorization.

3. User and permission model

User state	Can do	Cannot do until connected
Guest / local-only	Open local files, read offline, use recents, bookmarks/highlights, adjust reader settings, download/import files through Android picker.	Browse Google Drive, upload to Drive, cloud session sync.
Signed in, Drive not connected	All local features; account-level app profile if implemented.	Access Drive content.
Drive connected	Local features plus browse Drive, open cloud documents, upload, download, save	Access files not permitted by Google Drive or available to the connected account.

	supported edits, cloud recent workspaces.	
Drive authorization expired/revoked	Local files and downloaded copies remain available.	New Drive requests until reconnect.

3.1 First-run experience

- Welcome screen presents Open a file as the primary action and Connect Google Drive as a secondary optional action.
- Explain: 'You can read local files without an account. Connect Drive only when you want cloud files or transfers.'
- Do not show a Google consent screen on app launch.
- Ask for system file access only when a user picks a file or folder, using platform pickers rather than broad storage permission where possible.
- After first file open, offer a short tip explaining Recents and offline availability; do not force onboarding completion.

4. Mobile navigation and information architecture

Use a bottom navigation bar with four destinations: Home, Library, Activity, and Settings. The primary reader is a full-screen destination opened from any document. On small phones, avoid persistent sidebars and multi-column layouts.

Destination	Purpose	Key elements
Home	Resume reading and open files quickly.	Continue reading card, Recent files, Open file button, optional Drive connection card, offline status.
Library	Browse local and cloud content.	Segmented Local / Google Drive views; local picker; folders; sort/filter/search; downloaded-for-offline section.
Activity	Track transfers and recent cloud actions.	Uploads, downloads, scans, failures, retry/cancel, last sync time.
Settings	Control reading and account behavior.	Theme, font, reading width, line spacing, offline storage, Drive connection, privacy, help.
Reader	Distraction-free document reading.	Document title, back navigation, overflow actions, format-specific toolbar, content, find, bookmarks/highlights.

4.1 Navigation behavior

- Use Android system back behavior: close dialogs, then reader, then return to previous Library/Home state.
- Preserve each Library tab's scroll position and folder location while switching between Local and Drive.
- Use bottom sheets for actions and confirmations; use full-screen pages for folder browsing and reader settings where needed.
- Keep a visible provider label or icon for local vs Drive documents.
- Do not expose a desktop-style file tree permanently on phones; use a folder breadcrumb and back navigation.

5. UI design system

5.1 Visual direction

Token	Recommended value	Usage
Brand / primary	#4F46E5	Primary actions, selected state, links and progress.
Secondary accent	#6366F1	Secondary emphasis and interactive states.
Optional cyan	#06B6D4	Occasional cloud/status accent; use sparingly.
Light background	#F8FAFC	App background.
Surface	#FFFFFF	Cards, sheets and reader chrome.
Primary text	#0F172A	Titles and body UI.

Secondary text	#64748B	Metadata and helper copy.
Border	#E2E8F0	Separators and card outlines.
Success	#15803D	Saved, available offline, completed.
Warning	#B45309	Pending, conflict, reconnect required.
Error	#B91C1C	Failed transfer, permission issue.
Dark background	#0B1220	Dark theme background.
Dark surface	#111827	Dark theme cards and chrome.

5.2 Typography and spacing

- Use a native system sans-serif for interface labels; offer readable serif/sans options in the reader.
- UI text: 14–16 sp body, 12–13 sp metadata, 20–28 sp page titles; do not make essential actions smaller than 14 sp.
- Reader body: adjustable 16–24 sp default range, 1.45–1.8 line height, comfortable margins, and optional content width.
- Use an 8 dp spacing rhythm, 12–16 dp card padding, 16–20 dp screen gutters, and 12–18 dp corner radii.
- Touch targets should be at least 48 dp where practical; icons need labels or accessibility descriptions.

5.3 Component rules

- Use restrained borders and subtle elevation; avoid excessive gradients, glass effects, and oversized cards.
- Use consistent file-type icons and labels; do not rely on color alone to convey file type or status.
- Every async action must have loading, success, empty and error states.
- Dialogs should explain the consequence of the action; destructive actions require explicit confirmation.
- Support light, dark and system themes. Sepia is an optional reader theme.

6. Screen-by-screen UI specification

6.1 Welcome / first launch

Layout: brand mark and short value statement at top; primary Open a file button; secondary Connect Google Drive card; small 'Works offline for local files' note; no login gate.

```
[ReadSpace mark]
Your documents. One calm reading space.
Read local files offline. Connect Drive only when you need it.

[ Open a file ]              primary
[ Connect Google Drive ]     secondary

Local reading works without an account.
```

Interactions: Open a file launches Android document picker. Connect Google Drive opens a short permission explanation followed by Google OAuth. Dismissed onboarding remains dismissed.

6.2 Home

Top bar: greeting or 'Good evening', search icon, profile/Drive status icon. Main content: Continue reading card, Recent files list, quick actions, and a non-intrusive Drive connection prompt if not connected.

```
Home
[Search]                                    [Settings]

Continue reading
[ PDF icon ] DBMS Unit 3.pdf
Page 18 · Opened yesterday             [Resume]

Quick actions
[ Open file ] [ Browse Library ]

Recent
[icon] Transactions.md       Yesterday
[icon] CN Notes.pdf          Mon
[icon] Compiler Design.docx  Fri
```

```
Optional card: Connect Google Drive to browse cloud files.
```

6.3 Library — Local

- Header: Library title and search.
- Segmented switch: Local | Google Drive. Drive segment shows Connect Drive when not connected.
- Local view: Recent, On this device, Downloads, Available offline, and user-picked folders where supported.
- Provide sort by Name, Last opened, Modified, and Type; filter by supported formats.
- Use Android Storage Access Framework / document picker rather than requesting broad storage access by default.
- For a picked folder, show breadcrumbs, subfolders first, file rows with type icon, title, size/date and overflow menu.

6.4 Library — Google Drive

- If disconnected, show Connect Google Drive and explain that local reading remains available.
- If connected, show account avatar/email, root navigation, breadcrumbs, search, sort, and refresh.
- List folders and files lazily with pagination. Show cloud icon and offline/downloaded status separately.
- Do not load the entire Drive at once; fetch folder contents on demand.
- Offer Upload, Download, Make available offline, and supported file actions in an overflow menu.
- If Drive is offline, show cached metadata only where available and label it stale; do not imply cloud files can open offline unless cached/downloaded.

6.5 Reader screen

The reader uses a full-screen layout. Top app bar has back, truncated filename, source/status indicator and overflow actions. A minimal bottom/overflow toolbar provides Find, bookmark, highlight, text settings, download/offline and share/open-with where supported.

```
[Back]  DBMS Unit 3.pdf        [Cloud/Local] [⋮]
-------------------------------------------------
             document content
             page / text / tables
             ...
-------------------------------------------------
[Find] [Bookmark] [Aa] [More]
PDF: page count, page jump, zoom and fit-width.
Markdown: outline, links, code blocks, tables.
TXT: wrap, line numbers optional.
DOCX: semantic formatted preview; limited complex-layout fidelity.
```

- Persist scroll position or PDF page/zoom periodically and when leaving the reader.
- Restore position when the same resource is reopened, unless the user explicitly chooses Start from beginning.
- Reader controls should hide automatically while reading where appropriate, with tap-to-reveal behavior.
- HTML must be sanitized and sandboxed; external links open in a safe browser context after user action.
- For cloud files, show whether the current copy is cloud-only, downloaded, dirty, or pending upload.

6.6 Recent and Continue Reading

- Recent list is available locally without login.
- Cloud recent workspaces can sync to the user's account after sign-in; local recents remain device-specific unless user opts into metadata sync.
- Show last-opened time, document type, source and offline badge.
- Provide remove-from-recents separately from delete-file actions.
- Continue Reading resumes the last valid document and position; if unavailable, offer the next recent item.

6.7 Upload selected files

1. User opens a local folder or picks files and taps Upload.
2. If Drive is not connected, explain why connection is required and offer Connect Google Drive.
3. User chooses the destination Drive folder.
4. Review screen lists selected files, sizes, possible duplicate/name collision warnings and total size.
5. User confirms upload; progress appears in Activity and each item reports success/failure.
6. After success, offer Open in Drive folder. Never imply that a local file was moved; upload normally creates a cloud copy.

6.8 Upload new files

7. User selects a local folder and chooses a destination Drive folder or a saved folder mapping.
8. ReadSpace enumerates local files and compares them to the Drive destination using relative path/name plus size/modified time as hints; hash comparison may be used where feasible.
9. Show a review list classified as Likely new, Possible match, Changed, Name collision, Unsupported, or Cannot read.
10. User checks which files to upload. No upload begins before confirmation.
11. After upload, save mappings/Drive IDs and show per-file results.

6.9 Download / Make available offline

- Download selected: select Drive items, choose a device destination using the system picker when supported, then download with progress.
- Make available offline: download a copy into ReadSpace-managed app storage and list it in Available offline.
- For Google Docs/Sheets/Slides, ask for an export format such as PDF or DOCX where supported.
- Warn when the device lacks storage; allow cancel/retry.
- Show conflict choices for local filename collisions: Keep both, Replace after confirmation, or Skip.
- Deleting an offline copy must not delete the original cloud file.

6.10 Activity / Transfers

Item state	Visual behavior
Scanning	Spinner/progress and 'Checking local folder against Drive…'.
Needs review	Badge plus Open review button; no transfer has started.
Uploading / Downloading	Progress bar where available, bytes/count, Cancel action.
Completed	Success icon, destination and timestamp; allow Open destination.
Partial failure	Show succeeded and failed counts separately; Retry failed items.
Offline / paused	Explain that internet is needed for Drive operations; keep local reading available.
Conflict	Warning icon and explicit resolution options.

6.11 Settings and account

- Reading: font family, size, line height, margins/content width, page theme, screen-awake preference.
- Appearance: system/light/dark and optional sepia reader mode.
- Offline storage: cache size, downloaded files, clear cache, remove offline copies, storage usage.
- Google Drive: connected account, connection status, reconnect, disconnect, permission explanation.
- Privacy: local-only behavior explanation, clear local recents, clear local drafts, delete ReadSpace account where supported.
- About/help: supported formats, browser/app version, feedback, privacy policy.

7. Mobile functional requirements

ID	Requirement	Priority	Acceptance criteria
MOB-01	Local use without login.	P0	A fresh install can open and read a local file without account creation.
MOB-02	System document picker.	P0	User can select a supported file without broad storage permission where platform permits.
MOB-03	Offline recent list and position restore.	P0	Recently opened local files and last position persist across app restart when the URI/file remains accessible.
MOB-04	Supported format reader.	P0	PDF/TXT/Markdown/HTML/DOCX show appropriate viewer or clear unsupported error.
MOB-05	Optional Google sign-in/Drive consent.	P0	No consent screen appears unless user initiates a cloud action.

MOB-06	Browse Drive.	P0	Connected user can browse accessible folders with pagination and loading/error states.
MOB-07	Upload selected local files.	P0	Chosen files upload to chosen folder with progress/results.
MOB-08	Detect and upload new files.	P0	User reviews candidates and confirms before transfer.
MOB-09	Download selected Drive files.	P0	User can download/export to device or app-managed offline storage.
MOB-10	Make available offline.	P0	Downloaded copy opens with no network and has an offline badge.
MOB-11	Bookmarks/highlights.	P1	Supported document positions and annotations persist locally.
MOB-12	Account-level cloud recents.	P1	Drive workspaces and reading metadata restore after sign-in on another device where content is accessible.
MOB-13	Transfer activity and retry.	P0	Partial failures are visible and safe to retry.
MOB-14	Conflict handling.	P0	App never silently overwrites a changed cloud file.
MOB-15	Dark and reader themes.	P0	All primary screens and reader controls support themes.
MOB-16	Accessibility.	P0	TalkBack/VoiceOver labels, scalable text, contrast and keyboard/switch access where applicable.

8. Offline behavior specification

Feature	Offline behavior
Open local file	Works when OS URI/file access remains available.
Read previously downloaded file	Works offline.
Read cloud-only file not cached	Unavailable; show Connect to internet or Download for offline.
Browse Drive	Cannot refresh remotely; may show stale cached listing with an explicit stale label.
Upload/download	Queued only if the user chooses; transfer begins when online. Never claim success before server confirmation.
Bookmarks/highlights	Save locally immediately; optional cloud metadata sync when signed in and online.
Session restore	Local state restores; cloud resources may require reconnection or revalidation.
Edit local TXT/MD/HTML	Save locally where provider permissions allow; recover draft if saving fails.
Edit cloud document offline	Allow only if an offline copy/editor path is explicitly supported; mark as a local draft pending upload and detect conflicts on reconnect.

9. Technical architecture

Recommended mobile client: React Native with Expo development builds if the project benefits from the existing React/TypeScript skills. Use native modules or platform APIs where required for document picking, filesystem access, secure storage and robust PDF viewing. A fully native Kotlin implementation is also valid if Android-only support is the priority, but would duplicate more of the existing React frontend logic.

```
ReadSpace Mobile (React Native / TypeScript)
  ├── UI: Home, Library, Reader, Activity, Settings
```

```
  ├── LocalDocumentProvider
  ├── GoogleDriveProvider (API client; no client secret)
  ├── Reader renderer adapters
  ├── Local persistence: SQLite/AsyncStorage + filesystem cache
  ├── Secure credential storage: Keychain/Keystore via secure storage
  └── API client
          |
          v
Vercel backend
  ├── Google OAuth callbacks / session endpoints
  ├── Drive operations / token refresh
  ├── MongoDB repositories
  ├── Workspace/session APIs
  └── Transfer orchestration and audit metadata
          |                          |
          v                          v
    MongoDB Atlas              Google Drive API
```

9.1 Suggested mobile stack

Concern	Recommendation	Notes
Mobile UI	React Native + TypeScript	Expo development build recommended if native file access/PDF needs are supported by chosen packages.
Navigation	React Navigation or Expo Router	Bottom tabs plus stacked reader/folder screens.
Local metadata	SQLite	Reliable structured recent files, bookmarks, reading positions and transfer queue.
Small preferences	AsyncStorage	Theme and simple settings only.
Secrets/credentials	OS Keychain/Keystore-backed secure storage	Never store refresh tokens in ordinary AsyncStorage.
Local files	Android Storage Access Framework / document picker; iOS document picker	Persist URI permissions where platform allows; handle revoked permissions.
PDF	A maintained native PDF viewer	Evaluate offline rendering, text search, page navigation, selection and zoom.
Markdown/TXT/HTML	Renderer with sanitization	Sanitize HTML and external links; no unsafe script execution.
DOCX	Conversion/rendering library or supported server conversion only when opted in	Prefer on-device parsing where reliable; disclose any upload to a conversion service.
Cloud auth	Backend OAuth authorization-code flow; app deep link/universal link callback as appropriate	Use PKCE/state and secure app session; no client secret in mobile binary.
Networking	Typed API client with retries	Bounded retries, cancellation, token/session error handling.
Testing	Jest/Vitest-compatible tests, React Native Testing Library, Maestro/Detox	Test offline and permission-revocation scenarios.

9.2 Authentication and secrets on mobile

- Local reading must work without app account session.
- When a user taps Connect Drive, launch the supported OAuth authorization flow using the system browser or approved provider SDK, with PKCE/state as appropriate.
- Do not embed GOOGLE_CLIENT_SECRET in the app. Mobile applications are public clients and cannot keep a client secret confidential.
- The backend owns confidential OAuth code exchange and refresh-token handling when using the server-side flow.
- Use a short-lived app session token or secure cookie strategy suitable for mobile; store credentials only in Keychain/Keystore-backed storage.
- Do not ask for broad Google Drive access until the user explicitly chooses Drive integration; explain the scope and purpose.
- When a user disconnects Drive, stop cloud requests and clear/revoke connection credentials according to the product policy. Keep local downloaded files unless the user separately deletes them.

10. Mobile data model

Entity	Fields	Storage
LocalDocument	localId, displayName, uri/bookmark, mimeType, size, lastOpenedAt, lastPosition, permissionState	SQLite; URI permission where supported by OS
RecentItem	resourceKey, provider, title, lastOpenedAt, viewState, offlineState	SQLite locally; optional account sync for cloud items
Bookmark	resourceKey, anchor/page/offset, label, createdAt	SQLite local; optional cloud metadata sync
Highlight	resourceKey, quote/anchor/page range, color, note, createdAt	SQLite local; content/anchor limitations vary by format
OfflineCopy	driveFileId, localCachePath, mimeType, downloadedAt, remoteVersion/modifiedTime, size	SQLite metadata + app-managed filesystem
TransferQueueItem	id, direction, source/destination, status, progress, retryCount, errorCode	SQLite local; backend transfer metadata where appropriate
DriveConnection	connection status, account email/subject, scopes, lastValidatedAt	Safe metadata locally; refresh token stays backend-side
WorkspaceSession	workspaceKey, tabs, active resource, reader positions, lastOpenedAt	SQLite local; optional cloud session state in MongoDB via backend

11. Mobile-specific API needs

Endpoint (illustrative)	Purpose
GET /api/auth/mobile/start	Create a state/PKCE-bound OAuth start flow.
POST /api/auth/mobile/exchange	Exchange a valid code or complete the secure mobile session handoff.
GET /api/auth/me	Return safe profile/session status.
POST /api/auth/logout	Invalidate app session.
GET /api/drive/status	Check connection and reauthorization state.
GET /api/drive/files?parentId=...	Paginated folder contents.
GET /api/drive/files/:id/metadata	Get name, MIME type, size and version metadata.
POST /api/drive/files/upload-session	Create an upload session or obtain a safe transfer plan.
POST /api/drive/files/:id/update	Save supported edits with conflict check.
POST /api/drive/files/:id/export	Export Google-native documents to supported format.
GET /api/workspaces/recent	Load account-level cloud recent workspaces.
PUT /api/workspaces/:id/session	Save cloud workspace tabs and positions.
POST /api/transfers/scan	Compare a local file manifest with a Drive folder.
POST /api/transfers/:id/retry	Retry failed items safely.

Mobile file bytes should generally move from the app to Google Drive using an appropriate authorized/resumable flow where feasible. Avoid sending large file bodies through Vercel functions if platform limits may be exceeded. Backend authorization and destination validation remain mandatory.

12. Mobile performance and storage

- Use lazy loading for file lists and paginated Drive folders.
- Render PDF pages on demand and release off-screen resources.
- Cache document metadata separately from document bytes.
- Set a configurable cache budget; show storage usage and allow clearing cache without deleting source files.
- Keep downloaded offline copies distinct from temporary render caches.

- Use background transfer only where platform restrictions and user-visible progress support it; otherwise keep transfer foreground and resumable.
- Use cancellation and retry for large downloads/uploads; preserve completed items after partial failure.
- Avoid scanning the entire device filesystem. Only scan locations the user explicitly selects or grants access to.

13. Security, privacy and platform requirements

- No login is required for local reading; do not collect a user's identity simply to open a local document.
- Use least-privilege permissions and platform file pickers; explain every permission in plain language.
- Encrypt sensitive app data at rest where practical; use Keychain/Keystore for local secrets.
- Keep OAuth client secret and refresh tokens out of the mobile binary, logs, analytics and crash reports.
- Apply the same account isolation checks on the backend for mobile and web clients.
- Sanitize HTML/Markdown-derived HTML and block script execution in document previews by default.
- External links require a user tap and open in a safe browser context.
- Offer disconnect, clear local recents, clear cache, remove offline copies and account deletion flows where applicable.
- Do not claim Google OAuth verification is unnecessary for public release; broad Drive scopes can require verification and applicable security assessment.
- Respect Android and iOS background task, file access, battery and storage policies.

14. Accessibility and quality

- TalkBack and VoiceOver labels for controls, file types, upload/download status, bookmarks and reader actions.
- Dynamic text sizing without clipping; minimum contrast for body text and status indicators.
- Support reduced motion and avoid animation as the only feedback.
- Touch targets around 48 dp/pt where practical.
- Reader controls accessible without gestures alone; provide labeled alternatives to swipe actions.
- Test small phones, large phones, tablets, portrait/landscape, system font scaling, dark theme and offline mode.

15. Mobile test plan

Scenario	Expected result
Fresh install, airplane mode	User opens local file picker and reads local document without login.
Open local PDF, close app, reopen	Recent entry appears and reader restores last page when URI remains accessible.
OS permission revoked	App shows unavailable state and prompts user to reselect; no crash.
Drive disconnected	Local Library works; Drive tab shows Connect Drive.
Drive connected, network lost	Downloaded files remain readable; cloud-only files show offline state.
Upload selected file	Destination is confirmed, progress shown, final status accurate.
Upload new files scan	Review list appears; nothing uploads until user confirms.
Duplicate names	App offers skip/rename/replace choices and never silently overwrites.
Download Drive PDF for offline use	Offline copy opens after disabling network.
Google-native file download	User is offered supported export format.
Refresh token invalid/revoked	Cloud actions request reconnect; local files remain available.
Two different accounts	Each account sees only its authorized Drive and its own account-level workspace metadata.
Large file transfer interrupted	Partial state is shown and retry does not create unintended duplicates.
Malicious HTML document	Scripts do not execute in preview; external links are controlled.
Accessibility	Core workflows pass TalkBack/VoiceOver review and dynamic text checks.

16. Mobile release roadmap

Phase	Scope	Deliverable
M0 — UX prototype	Screen flows, design tokens, navigation, reader wireframes and usability review.	Clickable prototype before backend integration.
M1 — Offline reader	Local file picker, PDF/TXT/MD reader, recents, reading position, themes.	Useful without login or network.
M2 — Local library	Recent/search/sort, bookmarks, local metadata database, recoverable drafts where supported.	Stable local-first experience.
M3 — Optional account + Drive	Google authorization, Drive browsing, open cloud files, account connection status.	Cloud reading when online.
M4 — Transfers	Upload selected, upload-new-files review flow, download, make available offline.	Core cloud file operations.
M5 — Session sync + hardening	Cloud workspace metadata restore, conflict states, accessibility, performance, security review.	Production candidate.
M6 — iOS and advanced sync	iOS adaptation, background transfers where appropriate, Drive Changes API, optional advanced sync.	Expanded platform support.

17. Acceptance criteria for mobile MVP

- A new user can read a local file without signing in or connecting to the internet.
- Recent local documents and reading positions persist where platform file access remains valid.
- The app clearly distinguishes local files, downloaded offline copies and cloud-only documents.
- Users can opt in to Drive connection and browse their own accessible folders.
- Upload selected and upload new files include a destination, review, progress and per-file results.
- Users can download Drive files or export supported Google-native documents for offline reading.
- No client secret or refresh token appears in the mobile bundle or local plain-text storage.
- Users can disconnect Drive without losing access to unrelated local files.
- App works gracefully with revoked file permissions, revoked OAuth access, offline state, failed transfers and name collisions.
- Core flows pass device testing on target Android versions and a defined iOS support matrix if iOS is in scope.

18. Product decisions to finalize before implementation

Decision	Recommended default
Initial platform	Android first; keep shared React Native code and design compatible with iOS.
Offline cache policy	Explicit Make available offline; temporary reader cache is separate and clearable.
Login requirement	No login for local features; Google sign-in/Drive consent only on cloud intent.
Document editing	MVP: read all supported formats; edit local Markdown/TXT/HTML only if reliable on mobile; PDF/DOCX read-only.
Drive folder mapping	User chooses destination and can save a mapping; never infer silently.
Sync model	Manual scan and transfer first; continuous/background sync later.
Cross-device reading history	Sync cloud document metadata/position when signed in; local-only file history stays on device by default.
DOCX fidelity	Semantic content first; warn that complex layouts may differ.
Mobile download target	Support app-managed offline copies plus system destination picker where the OS allows it.

Appendix A — Design handoff checklist

- Create Figma frames for: Welcome, Home, Local Library, Drive Library disconnected, Drive Library connected, Folder contents, Reader PDF, Reader Markdown, Reader TXT, Upload review, Transfer progress, Download/export format, Recent workspaces, Settings, Drive reconnect, offline/empty/error states.
- Create components for: bottom navigation, file row, folder row, source badge, offline badge, connection card, search/filter bar, reader toolbar, progress row, review checklist, conflict dialog, empty state, permission recovery sheet.
- Prototype the complete no-login local journey before designing the Drive journey.
- Test the Drive consent prompt only after the user taps Connect Drive or a cloud-only action.
- Use realistic long filenames, nested folders, missing thumbnails, large files, no network, revoked access and accessibility font scaling in design QA.

Appendix B — Relationship to ReadSpace desktop/web

Mobile reuses product concepts and renderer behavior from the ReadSpace desktop/web PRD, but the UX is deliberately adapted for touch screens. The mobile app does not replicate the desktop Explorer sidebar and tab bar literally. It prioritizes full-screen reading, a simple Library, a compact reader toolbar, offline-first access and optional Drive workflows. Backend concepts such as MongoDB user/workspace metadata, Google OAuth, secure token handling, folder mapping, upload-new-files detection, transfer status and conflict checks are shared across web and mobile clients.

End of ReadSpace Mobile PRD + UI/UX + TRD v1.0